

Sanjay Chhetri

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U.S. Permanent Resident (Green Card Holder) — no sponsorship required

SUMMARY

Data scientist specializing in machine learning, NLP, and statistical modeling with experience developing analytical solutions across misinformation detection, behavioral research, and recommendation systems. Proficient in Python, R, SQL, and modern data science workflows including predictive modeling, experimentation, feature engineering, and data visualization.

TECHNICAL SKILLS

Programming Languages: Python, R, SQL

Machine Learning & AI: Scikit-learn, TensorFlow, PyTorch, NLP, Transformer Models, Generative AI Applications, Prompt Engineering, Feature Engineering, Regression, Classification, Model Evaluation

Data Science & Analytics: Pandas, NumPy, PySpark, Statistical Modeling, Hypothesis Testing, A/B Testing, Data Wrangling

Cloud, Deployment & MLOps: AWS, Docker, Databricks, MLflow, Hugging Face, Streamlit

Visualization & BI: Power BI, Tableau, Matplotlib, Seaborn, ggplot2

Databases & Development Tools: PostgreSQL, MySQL, Git, GitHub, Quarto, VS Code, Jupyter Notebook

PROJECTS

Tentacles of Misinformation

- Built NLP misinformation detection pipeline on 21k+ news articles using TF-IDF, SVM, and RoBERTa models (ROC-AUC: 0.870)
- Engineered psycholinguistic and behavioral features improving hybrid model robustness over NLP-only baselines
- Developed reproducible Quarto workflow and interactive dashboard for exploratory analysis and model evaluation

Emotion-Based Book Recommender

- Built emotion-aware recommendation engine using SBERT embeddings on the GoodBooks-10k dataset
- Improved Precision@10 by 71% and NDCG@10 by 113% over TF-IDF + KNN baselines
- Developed and deployed Streamlit application for personalized recommendation delivery

Teacher Assistant Dashboard

- Built analytics dashboard integrating attendance, grades, and behavior data for early-risk student identification
- Developed predictive intervention prototype achieving 84% accuracy in identifying at-risk students
- Automated behavioral trend visualization reducing manual tracking workload by an estimated 60%

EXPERIENCE

Applied Data Science Fellow | Springboard | Remote, US

May 2025 – Present

- Executed end-to-end data science projects spanning NLP, recommender systems, predictive analytics, and statistical modeling, from data acquisition and feature engineering through model evaluation and deployment.
- Designed and implemented analytical workflows using Python, R, SQL, Scikit-learn, and TensorFlow to preprocess, analyze, and model behavioral, textual, and recommendation-system datasets.
- Applied statistical analysis, machine learning, experimentation, and data visualization to develop reproducible, data-driven solutions, documenting methodologies and communicating technical findings through reports and interactive dashboards.

Mathematics Educator | Harford County Public Schools | Bel Air, Maryland

Feb 2025 – June 2026

- Delivered AP Calculus, Honors Pre-Calculus, and Algebra instruction to 100+ students using data-informed strategies
- Analyzed classroom performance trends contributing to an estimated 15–20% improvement in assignment completion
- Developed analytical tracking workflows reducing manual performance-monitoring time by approximately 30%

STEM Educator | Green Meadow Waldorf School | Spring Valley, New York

Jan 2022 – Jan 2025

- Delivered mathematics and science instruction across 6+ secondary courses including AP Calculus and AP Physics
- Designed project-based quantitative learning experiences increasing engagement across STEM classrooms

Graduate Research Assistant | Montclair State University | Montclair, New Jersey

Aug 2020 – Jan 2022

- Conducted behavioral research using statistical modeling, mediation analysis, and experimental design
- Analyzed behavioral datasets in R using regression modeling, hypothesis testing, and psychometric evaluation

EDUCATION

- Montclair State University — M.S. Psychological Sciences
- Montclair State University — Graduate Certificate in Advanced Quantitative Methods
- Ramapo College of New Jersey — B.A. Psychology
- Tribhuvan University — B.S. Mathematics & English Literature